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GREATER NOIDA, UTTAR PRADESH

CodeAlpha Virtual Internship Program

Submitted By

SHOORVEER PRATAP AGRAHARI

Enrollment No.: 25SCS1003004868

B.Tech – Computer Science & Engineering with AIML

10 August 2026 – 10 September 2026

AN INTERNSHIP PRESENTATION

C++ PROGRAMMING INTERNSHIP

Task 1: CGPA Calculator
Task 2: Login & Registration System

Candidate & Program Overview



Candidate Details

Student Name: SHOORVEER PRATAP AGRAHARI

Enrollment No.: 25SCS1003004868

Degree: B.Tech (Computer Science & Engineering with AIML)

Academic Batch: 2025–2029

University: IILM University, Greater Noida, U.P.



Internship Profile

Program Name: CodeAlpha Virtual Internship Program

Specialization Domain: C++ Programming

Duration: 1 Month (10 August 2026 – 10 September 2026)

Student ID: CA/DF1/242251

Organization: CodeAlpha

Organization Profile: CodeAlpha

CodeAlpha provided the Virtual Internship Program in C++ Programming, offering a structured and practical environment for project-based learning.



Program Structure: The internship used task based projects to apply C++ programming concepts in practical console applications.



Learning Focus: Programming fundamentals, problem solving, input validation, file handling and menu-driven applications.



Program Outcome: Practical exposure to converting requirements into working C++ programs, testing and documentation.

Problem Statement & Industry Context

The Industry Challenge

Students may understand programming concepts theoretically but need practical experience in converting requirements into reliable working applications.

The internship addressed this through two practical console applications that required calculations, validation, authentication logic and file handling.

The Solution Scope

Task 1 solves subject-wise CGPA calculation using grade points and credit hours with validation.

Task 2 implements registration, duplicate username detection, login verification and local credential storage.

Core Project Objectives



C++ Fundamentals

Strengthen variables, data types, operators, conditions, loops and functions through practical programming tasks.



Console Applications

Implement small console-based applications from defined requirements and produce clear user-facing outputs.



Validation & File Handling

Practice input validation, incorrect-input handling, duplicate-user checking and basic local file handling.



Testing & Documentation

Test normal and invalid cases, document the projects and present the completed work professionally.

Technologies & Tools



Programming

Language: C++

Console-based application
development



Development Tools

IDE: Visual Studio Code

Compiler: GCC / MinGW



Version Control

Git for source-code version control

GitHub for repository hosting and
sharing



File Handling

C++ file streams for local data
storage

Operating Platform: Windows

Application Architecture

1

Task 1 — CGPA Calculator

Input: courses, grade points and credit hours

2

Validation

Grade point must be 0–10; credit hours must be positive

3

Weighted Calculation

Grade Point × Credit Hours for every course

4

GPA / CGPA

Total weighted points ÷ total credit hours

5

Task 2 — Login & Registration

Register → duplicate check → users.txt; Login → credential verification

1

Input Layer

Subject details, grades, credits, username and password

2

Validation Layer

Checks values, invalid credentials and duplicate usernames

3

Processing Layer

CGPA calculation or authentication logic

4

Storage Layer

Local text-file storage for registered credentials

5

Output Layer

Results, status messages and validation feedback

Methodology Timeline (Phases 1 to 4)



Methodology & Outcomes (Phases 5 to 6)



Projects, Features & GitHub Evidence

1

Task 1 — CGPA Calculator

Uses struct Course + vector<Course>; calculates weighted points and GPA/CGPA with validation.

2

Task 2 — Login & Registration

Menu-driven Register / Login / Exit flow with duplicate username and invalid credential handling.

3

File Handling

Uses ifstream/ofstream with users.txt; the credential file is excluded through .gitignore.

4

Testing Evidence

Repository contains screenshots for CGPA input/output/validation and login registration/error scenarios.

5

GitHub Repository

Two task folders include main.cpp, README.md and screenshots; the repository is publicly hosted on GitHub.

Certification & Official Verification

Internship Title	C++ Programming Virtual Internship
Issuing Organization	CodeAlpha
Program Duration	10 August 2026 – 10 September 2026
Student Name	Shoorveer Pratap Agrahari
Student ID	CA/DF1/242251
Certificate Date	10 September 2026

Certification & Official Verification



Thank You!



Shoorveer Pratap
Agrahari

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School of Computer Science &
Engineering

IILM University, Greater Noida, U.P.

[GitHub Repository: github.com/shoorveer45/CodeAlpha_CPP_Internship](https://github.com/shoorveer45/CodeAlpha_CPP_Internship)